AI AND THE FUTURE OF MICROGRIDS

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RESEARCH QUESTION

- What are the challenges faced by modern microgrids and how can AI address these challenges?

MODERN CHALLENGES TO THE ELECTRIC GRID

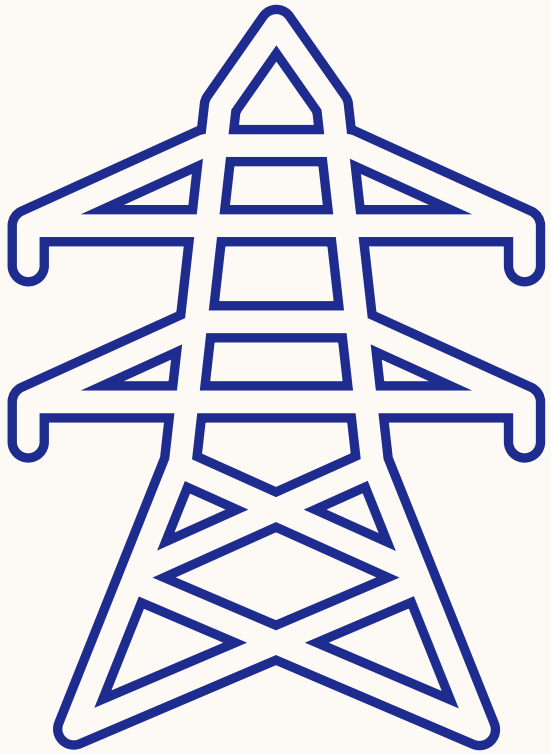
Reliability/Resiliency: Increased severe weather events

Integration of renewables: Intermittent energy sources challenge existing infrastructure

Increased demand: Data centers and electric vehicles

Energy storage: Integrating battery storage to assist in renewable generation

Transmission limitations: Moving energy from remote generation sites to densely populated areas



MICROGRIDS

MICRO GRIDS:

Microgrids are localized energy networks that can integrate with or operate islanded from the main electric grid.

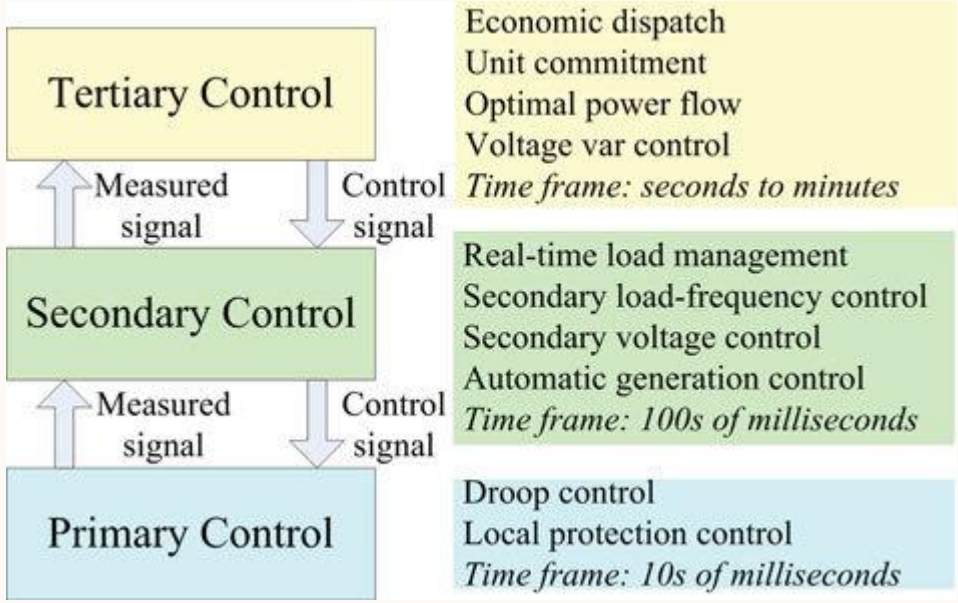
They can utilize local demand flexibilities to intelligently schedule energy generation and storage using DERs like batteries.

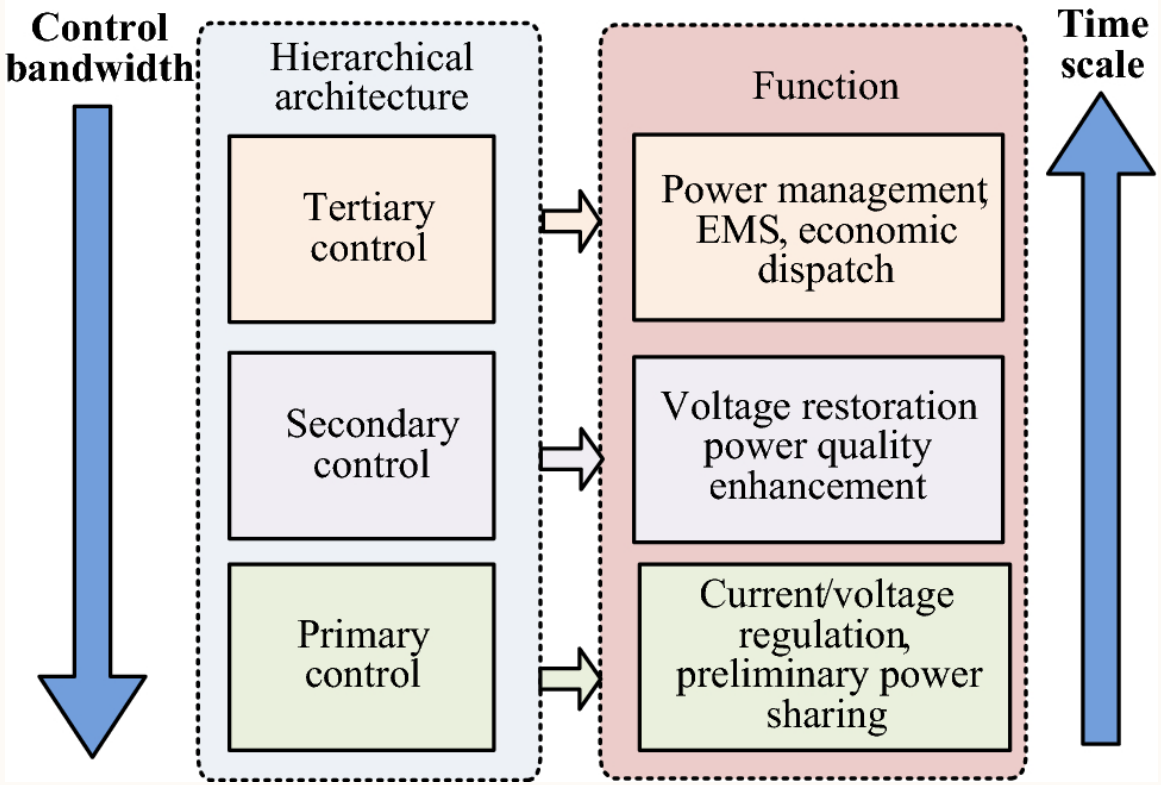
Address the problems of

- **Resiliency:** they can operate separate from the main grid powered by renewables and energy storage solutions
- **Improved Power Quality:** they can respond to flexible loads
- **Efficiency:** they can reduce losses incurred by transmission and distribution
- **Reducing CO2 Emissions:** they can improve and incentivize the use of rooftop solar, distributed generation, and battery storage

MICROGRID CONTROL

Microgrid control overview





MICROGRID CHALLENGES

- **Supply and Demand Prediction**
  - Predicting energy usage patterns as well as energy generation from intermittent renewables
- **Complex management and scheduling**
  - Energy storage and generation
  - Managing power flow within the grid
- **Grid integration**
  - Microgrids need to be able to act independently and integrate with the main grid or other microgrids
- **Fault response**
  - Microgrids have low inertia making them particularly susceptible to faults

ARTIFICIAL INTELLIGENCE

An Overview

ARTIFICIAL INTELLIGENCE

Supervised Learning

- AI that learns through making predictions on labeled data and updating its parameters to improve likelihood of outputting the correct answer

Semi-supervised Learning

- AI that learns from both labeled and unlabeled data or learns patterns from recreating a corrupted or partial example. Generative AI primarily works this way

Unsupervised Learning

- AI that learns patterns and trends in unlabeled data without known correct answers or human supervision

Reinforcement Learning

- AI that learns to take actions that optimize an objective function through a feedback loop where the AI explores the solution space and uses rewards to develop a strategy it then exploits

OVERCOMING CHALLENGES TO MODERN MICROGRIDS WITH AI

Framework for Using AI in Combating Climate Change

Topics		Mitigation		Adaptation and Resilience		Fundamentals
	Measurement	**Reduction**	**Removal**	**Hazard Forecasting**	**Vulnerability and Exposure management**	
Subtopics and examples	**Macro-level measurement** e.g., estimating remote carbon natural stock	**Reducing GHG emissions intensity** e.g., supply forecasting for solar energy	**Environmental removal** e.g., monitoring encroachment on forests and other natural reserves	**Projecting localized long-term trends** e.g., regionalized modeling of sea-level rise or extreme events such as wildfires and floods	**Managing crises** e.g., monitoring epidemics	**Climate research and modeling** e.g., modeling of economic and social transition
		Improving energy efficiency e.g., encouraging behavioral change			**Strengthening infrastructure** e.g., intelligent irrigation	**Climate finance** e.g., forecasting carbon prices
	Micro-level measurement e.g., calculating the carbon footprint of individual products	**Reducing greenhouse effects** e.g., accelerating aerosol and chemistry research	**Technological removal** e.g., assessing carbon-capture storage sites	**Building early warning systems** e.g., near-term prediction of extreme events such as cyclones	**Protecting populations** e.g., predicting large-scale migration patterns	**Education, nudging, and behavioral change** e.g., recommendations for climate-friendly consumption
					Preserving biodiversity e.g., identifying and counting species	

Uses for AI				
Gather, complete, and process data	**Strengthen planning and decision making**	**Optimize processes**	**Support collaborative ecosystems**	**Encourage climate-positive behaviors**
Satellite and IoT data Filling gaps in temporally and spatially sparse data	Policy and climate-risk analytics Modeling of higher-order effects Bionic management	Supply chain optimization Simulation environments	Vertical data sharing Enhanced communication tools	Climate-weighted suggestions Climate-friendly optimization functions

Sources: BCG project experience; Climate Change AI, "Tackling Climate Change with Machine Learning"; Global Partnership on AI, "Climate Change and AI: Recommendations for Government Action."

Note: GHG = greenhouse gas; IoT = Internet of Things.

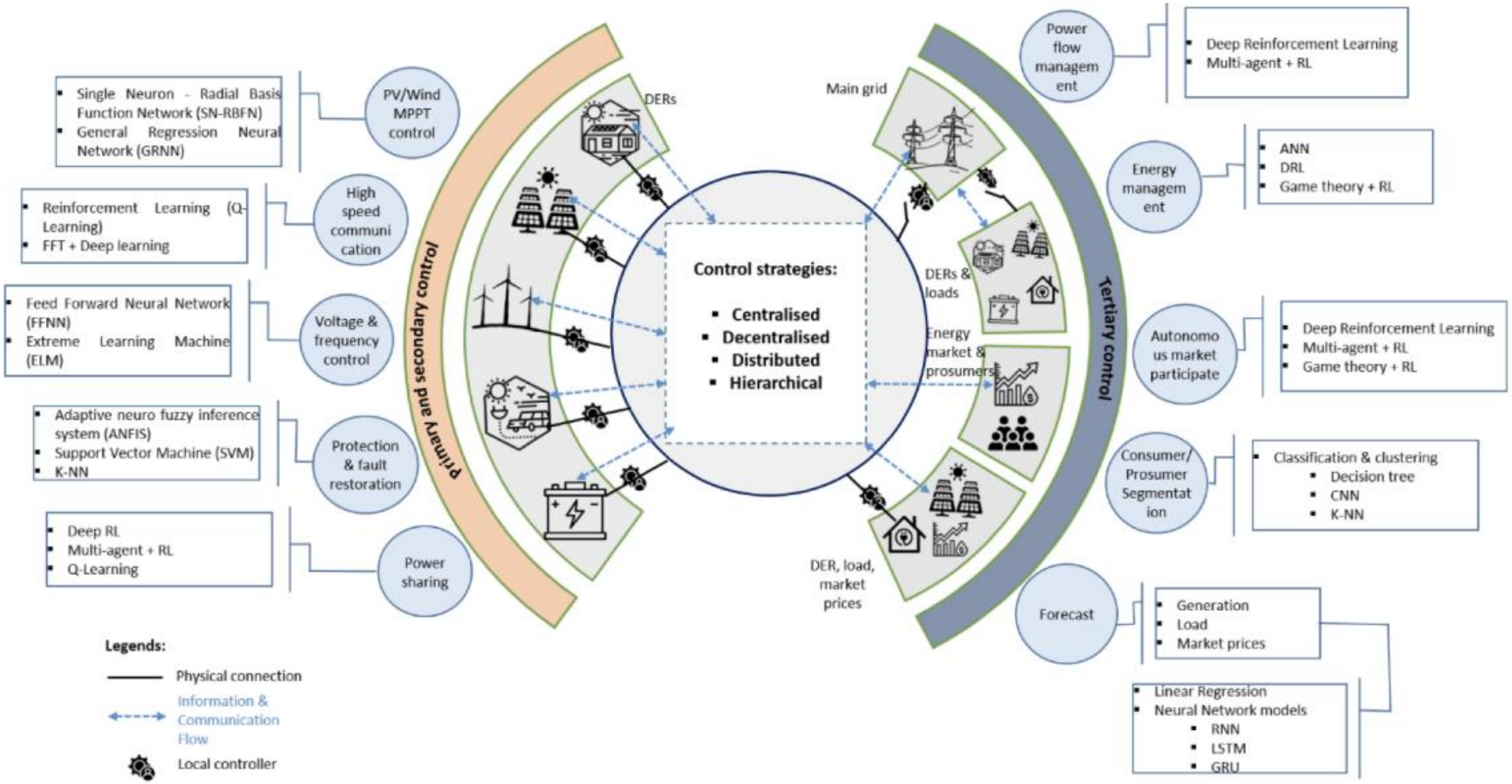


Fig. 3. Overview of AI techniques in various microgrid control and functionalities

PRIMARY CONTROL: IMMEDIATE STABILITY

Real Time Power Sharing

- Traditional model-based droop control lacks accuracy and adaptability
- Neural networks can improve droop control to improve active/reactive power sharing and voltage/frequency control

Maximum Power Point Tracking

- Single Neuron Radial Basis Function Networks and multilayer regression networks can improve reliability for tracking maximum power from PVs

Inertia Estimation & Control

- NN-based heuristic dynamic programming may help adapt to changing system conditions with minimal fluctuations

Droop is a control mechanism used in power generators to share the load between multiple generators

SECONDARY CONTROL: RESTORATION & OPTIMIZATION

Voltage & Frequency Deviation Control

- AI can help forecast load and demand near the source and Genetic Algorithms along with Neural Networks can quickly predict changing system dynamics

Communication Delay

- Delay minimization Q learning can reduce communication latency between hierarchical control components by up to 66%

Protection & Effective Fault Restoration

- Fast Fourier Transforms (FFTs) in conjunction with unsupervised anomaly detection can be used to quickly and accurately detect faults, and multi-agent reinforcement learning can work to control switches and segregate the fault and maintain power elsewhere

TERTIARY CONTROL: NETWORK MANAGEMENT

Optimizing economic dispatch of DERs

- Multi-agent reinforcement learning can coordinate battery charging and release as well as switches for distributed generation to achieve the lowest cost solution for load dispatch among participating DERs

Grid Integration

- NN-based EMS can be leveraged to minimize power imported from the main grid and maximize utilization of DERs dynamically

DER Market Participation

- Reinforcement learning can be used to optimize transaction decisions based on instantaneous information about supply and demand

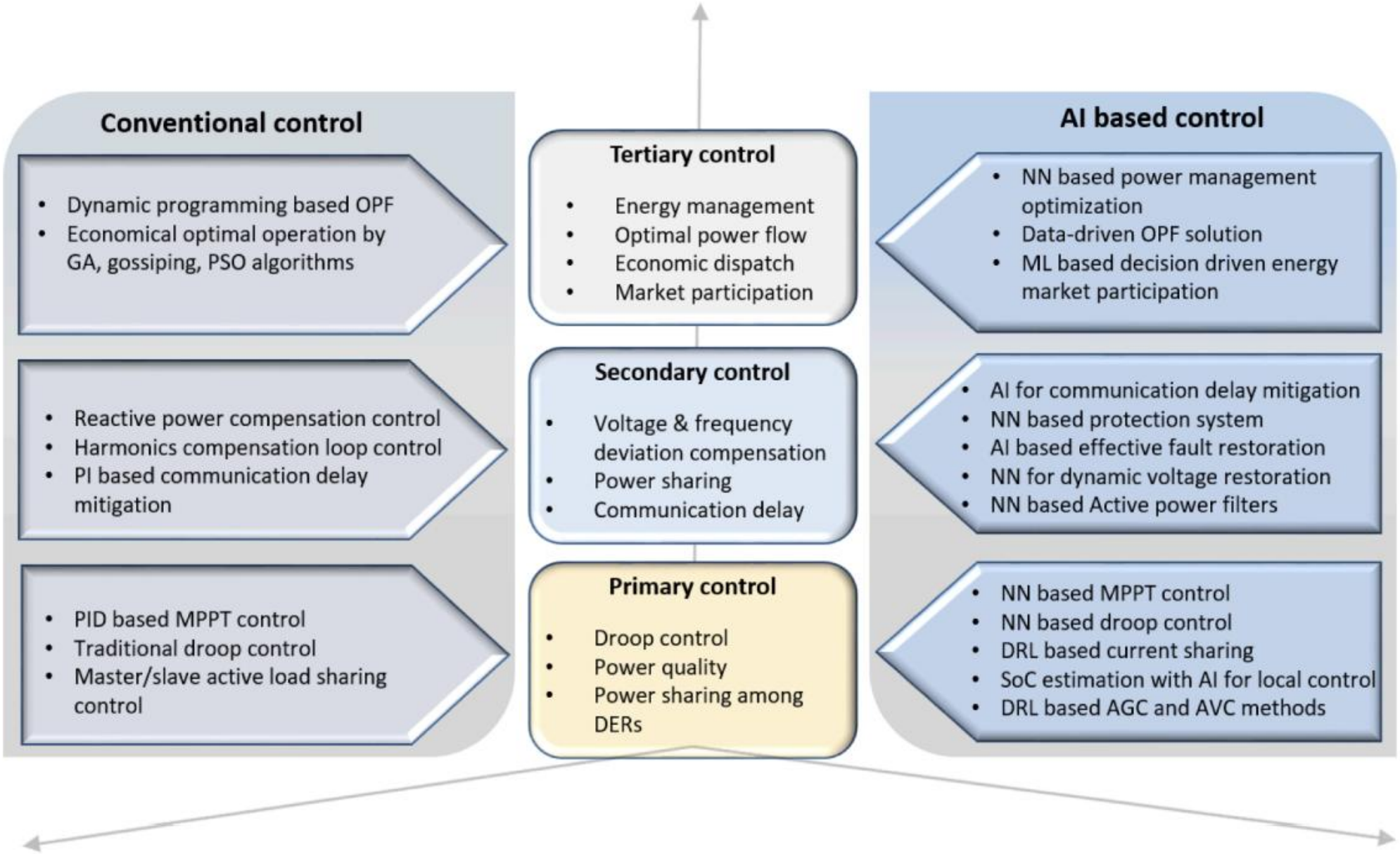


Fig. 7. Conventional and prospective AI-based control in the hierarchical control architecture

MICROGRID COORDINATION

Multiple microgrids can be networked together through switches to improve coordinated resiliency

- Reinforcement learning methods can be applied to open and close switches in response to grid events

Improved forecasting of renewable generation and load demand

- Microgrids can make more informed decisions regarding energy dispatch and trading

Neural Network based EMS can schedule across microgrids spanning multiple regions experiencing different weather patterns

- Distributing generation from a sunny area elsewhere and then redirecting as the cloud cover moves across the microgrids

LLM APPLICATIONS

Enhanced Human-Machine Interfaces for Microgrid Management

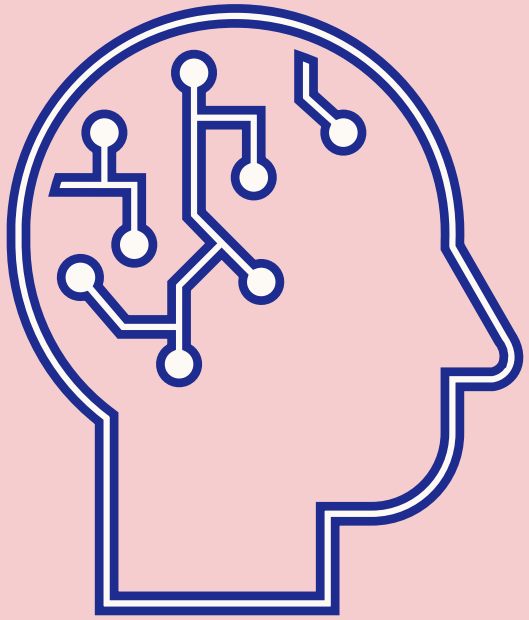
- LLMs could revolutionize how operators interact with microgrid Energy Management Systems (EMS)

Intelligent Diagnostics and Trouble-shooting

- LLMs could analyze system logs, alarm notifications, and real-time data to diagnose potential faults and suggest troubleshooting steps

Synthesis of Complex Information for Networked Microgrids

- LLMs could help synthesize information from multiple microgrid EMSs, providing a high-level overview of the entire network's status, identifying potential interdependencies, and suggesting coordinated control actions



ETHICAL CONSIDERATIONS & CHALLENGES

ETHICAL CONSIDERATIONS

Trustworthiness

- The consequences of grid failures can be severe, so in order for AI to be adopted, it must be proven to be trustworthy

Explainability

- Model interpretability: understand the mechanism of internal function
- Post-hoc explanation: provide insights based on inputs & outputs

Bias & Access

- Advanced microgrids may be trained for/accessed in high income areas and the cost of standard grid maintenance may fall on poorer consumers

Human Control

- AI should be able to be dialed back from full automation to augment or assist human decision makers if need be

CHALLENGES

Data

- There is a lack of data available for various problems such as inertia estimation in distribution networks. Additionally, as the number of DERs increases, old datasets may go "stale". Simulations may be a more appropriate method of training in many cases.

Real world testing

- Most of the papers on applying AI to microgrids work in simulations and have not been tested in real life environments. Thorough analysis must be done before applying any of these in the real world, yet it's difficult to verify without real implementation

Security

- All AI solutions must be deployed with consideration to cyber attacks and possible impacts on national energy security

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THANK YOU